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"Automated Kidney Lesion Classification from CT Images Using Convolutional Neural Networks with Explainability"

Research project report submitted by

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Under the guidance of

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CERTIFICATE OF ORIGINALITY

This is to certify that **Mahika Santosh Gupta**, a student of **Master of Science (Data Science & Artificial Intelligence),** has worked for the Research Project titled, **"Automated Kidney Lesion Classification from CT Images Using Convolutional Neural Networks with Explainability",** under my guidance for the degree of **Master of Science (Data Science and Artificial Intelligence),** Mithibai College (Empowered Autonomous) affiliated under University of Mumbai.

I further certify that the above said work was duly approved by me and is the outcome of candidate's individual efforts and study. The dissertation has not been previously used for the Degree of **Master of Science (Data Science & Artificial Intelligence)** of University of Mumbai or any other University of the examining body.

Certified by

____

Dr. Jayasree Ravi

DECLARATION

I **Mahika Santosh Gupta**, Student of **Master of Science (Data Science & Artificial Intelligence)**, **Semester IV**, Academic Year 2025-2026, wish to state that the work embodied in this report titled, **"Automated Kidney Lesion Classification from CT Images Using Convolutional Neural Networks with Explainability"** forms my own contribution to the research work carried out under the guidance of **Dr. Jayasree Ravi**. This work has not been submitted for any other degree of this or any other University. The information submitted is true and original to the best of my knowledge.

Mahika Gupta, L021

Master of Science

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CERTIFICATE

This is to certify that **Mahika Santosh Gupta**, L021 of Master of Science (Data Science and Artificial Intelligence), Semester IV, Academic Year 2025-2026 has successfully completed the project on **"Automated Kidney Lesion Classification from CT Images Using Convolutional Neural Networks with Explainability"** under the guidance of **Dr. Jayasree Ravi**.

Project Guide	Head of Department	External Examiner
(**Dr. Jayasree Ravi**)	**(Dr. Ashish Gavande)**	

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Mahika Gupta

ABSTRACT

Kidney disease is a growing global health concern, affecting over 850 million people worldwide. The increasing prevalence of kidney-related conditions such as cysts, stones, and tumors — driven by lifestyle factors including unhealthy diets, sedentary habits, and chronic stress — has placed a significant burden on healthcare systems globally. Early and accurate diagnosis of kidney lesions from CT scan images is critical, as timely intervention dramatically improves patient outcomes. For instance, kidney cancer diagnosed at an early stage carries a five-year survival rate exceeding 93%, which drops to below 12% when detected at an advanced stage.

Traditional methods of CT scan interpretation are time-consuming, expertise-intensive, and subject to inter-observer variability — challenges that are particularly acute in resource-limited healthcare settings. This study presents a comprehensive Automated Kidney Lesion Classification System that leverages Convolutional Neural Networks (CNNs) with transfer learning to classify CT images into four categories: Normal, Cyst, Stone, and Tumor.

Four deep learning models were implemented and comparatively evaluated under identical training conditions: a custom CNN baseline, MobileNetV2, EfficientNetB0, and ResNet50. The dataset comprised 5,708 CT images sourced from the publicly available CT Kidney Dataset on Kaggle. ResNet50 emerged as the best-performing model, achieving an accuracy of 81.04%, a precision of 0.865, a recall of 0.810, and an F1-score of 0.819 on the validation set. EfficientNetB0, by contrast, achieved only 20.23% accuracy, highlighting the critical importance of domain-specific model evaluation.

Beyond classification accuracy, the system incorporates Gradient-weighted Class Activation Mapping (Grad-CAM) to generate visual explanations that highlight the regions of CT images most influential to the model's predictions. A confidence-aware risk stratification framework categorizes predictions into Low, Moderate, and High/Uncertain risk levels to guide clinical decision-making. The system also features an automated two-page PDF diagnostic report generator and an AI-powered clinical assistant that provides condition-specific information and recommended next steps.

This work demonstrates that a clinically oriented AI pipeline — combining accurate classification, explainability, uncertainty quantification, automated reporting, and interactive assistance — offers a meaningful and practical approach to supporting radiologists and clinicians in kidney lesion diagnosis, particularly in settings with limited specialist availability.

Keywords

Kidney Lesion Classification, CNN, Transfer Learning, Grad-CAM, Explainable AI, CT Imaging, ResNet50, MobileNetV2, EfficientNetB0, Clinical Decision Support, Medical Image Analysis, Deep Learning

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Kidney disease is a significant health problem worldwide, affecting over 850 million people globally. The kidneys play a critical role in filtering waste products, balancing fluids, and regulating essential body functions. When kidney health deteriorates, the consequences can be severe and life-threatening. The three primary types of kidney lesions detectable on CT scans — cysts, stones, and tumors — can have vastly different clinical outcomes depending on when and how accurately they are diagnosed.

Among these conditions, kidney cancer (a form of kidney tumor) represents one of the most time-sensitive diagnoses in oncology. When detected early, the five-year survival rate exceeds 93%. However, if diagnosis is delayed until the disease reaches an advanced stage, the survival rate plummets to below 12%. This stark contrast underscores the immense clinical value of early, accurate detection.

CT (Computed Tomography) scans are the gold standard imaging modality for kidney lesion detection. However, the manual interpretation of CT scans is a time-consuming, expertise-intensive process. Radiologists must possess deep domain knowledge to accurately identify and differentiate between conditions that can appear visually similar. In high-volume healthcare environments and resource-limited settings, the growing demand for CT scan analysis outstrips the availability of qualified specialists, introducing delays and increasing the risk of diagnostic errors.

The advent of deep learning — and Convolutional Neural Networks (CNNs) in particular — has opened new possibilities for automated medical image analysis. CNNs are architecturally well-suited to image classification tasks: they learn hierarchical feature representations directly from pixel data, progressively identifying edges, textures, shapes, and complex anatomical structures without requiring hand-crafted feature engineering. Pretrained CNN models, developed on massive image datasets such as ImageNet, can be fine-tuned on domain-specific medical imaging data through a process known as transfer learning, enabling high-performance classification even with relatively limited labeled training data.

Despite significant advances in applying deep learning to medical imaging, a critical gap persists between research accuracy and clinical utility. Most existing systems focus exclusively on maximizing classification accuracy while neglecting the interpretability, uncertainty quantification, and clinical workflow integration that are essential for real-world deployment. This project addresses these gaps by developing a comprehensive, end-to-end Automated Kidney Lesion Classification System that not only classifies CT images with high accuracy but also explains its predictions, quantifies its confidence, and supports clinical decision-making through automated reporting and an AI-powered assistant.

1.2 Problem Statement

The automated classification of kidney lesions from CT images presents several interrelated technical and clinical challenges that existing systems have not fully addressed:

- Kidney lesions are often small, subtle, and difficult to distinguish visually — even for experienced radiologists — due to their resemblance to surrounding anatomical structures such as blood vessels and normal kidney tissue.
- Different lesion types (cysts, stones, tumors) can exhibit overlapping visual characteristics on CT scans, making precise differentiation challenging.
- Most existing deep learning classification systems operate as black boxes, offering no explanation of how or why a particular prediction was made. This lack of transparency is a significant barrier to clinical adoption.
- In healthcare settings with limited resources and high patient volumes, the need for scalable, efficient, and explainable diagnostic support tools is particularly acute.
- Errors in kidney lesion classification are not merely technical failures — they carry direct consequences for patient health, potentially resulting in missed diagnoses, delayed treatment, or unnecessary interventions.

This project aims to address these challenges by building a system that combines accurate deep learning-based classification with explainability (via Grad-CAM), confidence-aware risk stratification, automated diagnostic report generation, and an AI clinical assistant — delivering a complete and clinically oriented solution.

1.3 Objectives of the Study

The study aims to achieve the following specific objectives:

1. Build an end-to-end automated system capable of classifying kidney CT images into four categories: Normal, Cyst, Stone, and Tumor, with high accuracy and clinical reliability.
2. Implement and comparatively evaluate four CNN-based architectures — a custom baseline CNN, MobileNetV2, EfficientNetB0, and ResNet50 — under identical training conditions on the same dataset to enable a fair and rigorous model comparison.
3. Apply Grad-CAM (Gradient-weighted Class Activation Mapping) to generate visual heatmaps that highlight the anatomical regions most influential to the model's classification decisions, supporting radiologist verification.
4. Develop a confidence-aware risk stratification framework that translates raw model probability scores into clinically actionable risk levels (Low, Moderate, and High/Uncertain) to guide next steps.
5. Design and implement an automated two-page PDF diagnostic report generator that summarizes the system's findings in a format accessible to both clinicians and patients.
6. Integrate an AI-powered clinical assistant that provides condition-specific information, symptom awareness guidance, and recommended next steps based on the model's prediction.
7. Evaluate system performance rigorously using accuracy, precision, recall, F1-score, and confusion matrix analysis, with macro-averaged metrics to ensure fairness across imbalanced classes.

1.4 Scope of the Study

This study focuses on the binary-to-multi-class classification of kidney CT scan images into four categories: Normal, Cyst, Stone, and Tumor. The system is built using structured image data from a publicly available CT imaging dataset sourced from Kaggle.

The study was implemented using Python, leveraging deep learning frameworks including TensorFlow and Keras, alongside supporting libraries such as NumPy, Pandas, Matplotlib, and OpenCV. The system includes a Streamlit-based interactive web interface that allows users to upload CT images and receive real-time classification results, Grad-CAM visualizations, confidence scores, risk stratification outputs, downloadable PDF reports, and AI assistant responses.

The scope is intentionally focused on single-image CT scan classification and does not extend to video-based analysis, multi-modal data fusion (e.g., combining imaging with laboratory results or electronic health records), or longitudinal patient monitoring. The system is designed as a clinical decision support tool and does not function as a replacement for radiologist or physician judgment.

1.5 Significance of the Study

This study makes several meaningful contributions to the intersection of artificial intelligence and medical imaging:

First, it provides a rigorous, head-to-head comparison of four CNN architectures on the same kidney CT dataset under identical training conditions — a methodological standard that is absent from much of the existing literature, where models are often evaluated in isolation or on different datasets.

Second, the integration of Grad-CAM explainability directly addresses the black-box criticism that has limited clinical adoption of AI diagnostic tools. By visually demonstrating where the model focuses its attention, the system enables radiologists to verify, challenge, or build confidence in the model's predictions.

Third, the confidence-aware risk stratification framework introduces a clinically practical dimension to the system's output, acknowledging that model uncertainty is medically relevant and should inform how predictions are acted upon.

Fourth, the combination of automated reporting and an AI clinical assistant extends the system's utility beyond specialist clinicians to include general practitioners and patients — particularly valuable in underserved or resource-limited healthcare environments where specialist access is constrained.

Overall, this research advances the design principles for clinically integrated AI diagnostic systems, demonstrating that accuracy alone is insufficient and that explainability, uncertainty awareness, and clinical workflow integration are equally essential components of a trustworthy medical AI tool.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The application of deep learning to medical image analysis has grown substantially over the past decade, with kidney lesion classification emerging as an active research area owing to its significant clinical implications. Researchers have explored a wide range of CNN architectures, transfer learning strategies, data augmentation techniques, and explainability methods in pursuit of systems that are both accurate and clinically trustworthy. This chapter reviews key studies in this domain, identifies methodological strengths and limitations, and situates the present work within the broader research landscape.

2.2 Review of Existing Studies

Subedi et al. (2023) conducted a comparative study of ResNet50, VGG16, and a Vision Transformer on the CT Kidney Dataset from Kaggle. The Vision Transformer achieved an impressive accuracy of 99.64% on a 90:10 train-test split. However, this study did not incorporate any form of explainability, nor did it include uncertainty quantification or clinical decision support features. The absence of external validation on independent datasets also raises concerns about the generalizability of the reported accuracy.

Pande and Agarwal (2024) applied YOLOv8n-cls to kidney abnormality detection, achieving an accuracy of 82.52%. While YOLO-based architectures offer computational efficiency and are well-suited to real-time object detection tasks, they are less effective at extracting the deep hierarchical features required for nuanced lesion texture and density analysis. The study did not include explainability mechanisms or clinical support features.

Mehedi et al. (2022) demonstrated that combining segmentation (via UNet) with classification (via VGG16) can push accuracy to 99.48%. This segmentation-first approach provides richer spatial information, but it critically depends on pixel-level annotations — a form of ground truth labeling that is prohibitively expensive and rarely available in real clinical datasets.

Pimpalkar et al. (2025) achieved a remarkable accuracy of 99.96% using a modified InceptionV3 model with extensive data augmentation. However, concerns about dataset-specific overfitting are significant — results obtained on a single dataset without external validation are difficult to interpret as a reliable indicator of real-world performance.

Majid et al. (2023) applied DenseNet-121 to kidney tumor classification, achieving 98.22% accuracy. DenseNet's dense connectivity pattern enables effective feature reuse, contributing to strong performance. However, the study lacked explainability components and clinical decision support integration.

Ahmed et al. (2024) used VGG16 with Grad-CAM on KUB X-ray images for kidney stone identification, achieving 97.41% accuracy. This study is the closest in spirit to the present work in its use of gradient-based explainability. However, it was applied to X-ray rather than CT imaging, did not include confidence-based risk stratification, and lacked clinical recommendation features.

Gong and Kan (2021) explored segmentation and classification of renal tumors using CNN-based approaches on CT data. While their segmentation pipeline provided detailed spatial information, the computational overhead and annotation requirements limit practical scalability.

Alzu'bi (2022) introduced a custom dataset (KAUH) for kidney tumor detection and applied a 2D-CNN, achieving 97.70% accuracy. The use of an independently collected clinical dataset is a methodological strength, though the absence of explainability features limits clinical interpretability.

Asif et al. (2024) explored the fusion of multiple deep learning models for kidney stone detection from CT images, demonstrating that ensemble approaches can improve robustness. However, the study focused narrowly on stone detection rather than the full multi-class lesion classification problem.

Kia et al. (2024) investigated the fusion of VGG16, MobileNet, EfficientNet, AlexNet, and ResNet50 for brain tumor identification, providing valuable insights into cross-architecture comparative evaluation that inform the model selection approach adopted in the present study.

2.3 Research Gap

A synthesis of the existing literature reveals several recurring methodological limitations that motivate the present study:

- The majority of studies report accuracy as their primary or sole evaluation metric, which is insufficient — particularly on imbalanced datasets — and does not capture clinically critical performance characteristics such as false negative rates.
- Most existing systems operate as classification-only pipelines and do not incorporate explainability mechanisms. The absence of visual explanations makes it difficult for clinicians to verify the basis of model predictions, limiting trustworthiness and clinical adoption.
- Uncertainty quantification is largely absent from the literature. Most systems provide point predictions without communicating the model's confidence, leaving clinicians without crucial information about when to seek additional expert review.
- Clinical decision support features — including condition-specific guidance, recommended next steps, and automated report generation — are rarely integrated into existing classification systems.
- Many studies validate their models exclusively on the training dataset's test split, without external validation on independent clinical datasets, raising legitimate concerns about generalizability.
- High accuracy figures reported in some studies (99%+) may reflect data leakage, favourable train-test splits, or dataset-specific overfitting rather than genuine generalization capability.

The present study addresses these gaps by developing a system that integrates accurate CNN-based classification with Grad-CAM explainability, confidence-aware risk stratification, automated diagnostic reporting, and an AI clinical assistant — delivering a comprehensive, clinically oriented solution that goes significantly beyond classification accuracy alone.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Objective

The primary objective of this research is to design, implement, and evaluate a complete automated kidney lesion classification system using CNNs with transfer learning, supplemented by Grad-CAM explainability, confidence-based risk stratification, automated reporting, and an AI clinical assistant. The system is evaluated on a publicly available CT kidney dataset and benchmarked against existing methods in the literature.

3.2 Research Design

This study adopts an experimental machine learning research design. Four CNN architectures are trained and evaluated under identical conditions on the same dataset, enabling rigorous, fair model comparison. The final system integrates the best-performing model into a full clinical support pipeline accessible through a web-based interface.

3.3 Dataset

The project uses the CT Kidney Dataset originally shared on Kaggle by Islam et al. (2022). The full dataset contains 12,446 images; for this project, a balanced subset of 5,708 CT images was used to ensure fair class representation during training and evaluation.

Characteristic	Detail
Source	Kaggle – CT Kidney Dataset (Islam et al., 2022)
Total Images Used	5,708 CT images
Classes	Normal, Cyst, Stone, Tumor
Cyst	1,500 images
Normal	1,500 images
Tumor	1,500 images
Stone	1,208 images (minority class)
Training Set	4,559 images
Validation Set	1,149 images
Image Format	PNG / JPEG

Table 1: Dataset Overview

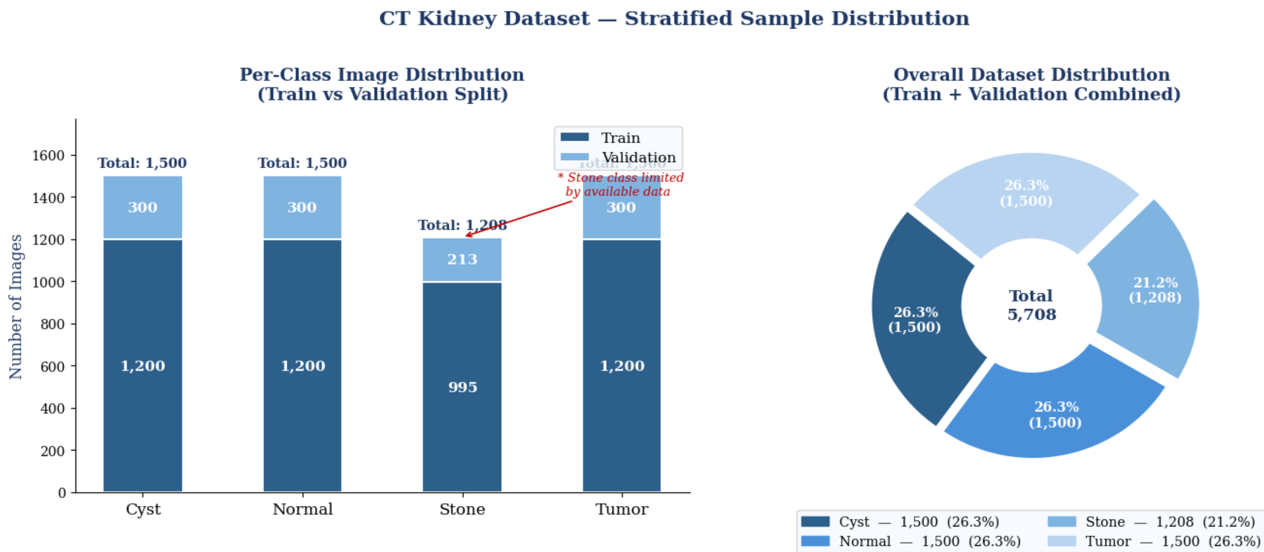


Figure 1: Distribution of Dataset Classes (Total = 5,708 Images)

The Stone class represents only 11.1% of the total dataset. To address this class imbalance, class weights were applied during training and additional oversampling of Stone images was performed. The validation set was kept entirely separate from training data, with no augmentation applied to validation images to ensure unbiased performance evaluation.

3.4 Preprocessing and Augmentation

All images were resized to 224 × 224 pixels and normalized by dividing pixel values by 255 to bring them into the [0, 1] range. Data augmentation was applied exclusively to the training set using Keras's ImageDataGenerator to improve model generalization and reduce overfitting.

Augmentation Parameter	Value / Configuration
Resize	224 × 224 px (bilinear interpolation)
Normalization	1/255.0 → [0, 1]
Rotation	±15°
Width / Height Shift	±5%
Zoom	±10%
Horizontal Flip	Enabled
Vertical Flip	Disabled (anatomical constraint)
Fill Mode	Nearest neighbour

Table 2: Training Augmentation Parameters

Vertical flipping was deliberately disabled as inverted kidney images are not anatomically realistic and would introduce spurious training signal. The Stone class imbalance was additionally addressed through class-weight assignment in the model's loss function.

3.5 System Architecture

The system is structured as a sequential end-to-end pipeline, from CT image upload through to diagnostic report generation and AI-assisted consultation. The pipeline comprises seven stages:

8. CT scan image upload (PNG/JPEG, 224×224 px)
9. Preprocessing: resize, normalize, augment (training only)
10. Comparative evaluation across four CNN architectures; best model selected
11. Classification output: Softmax probabilities for Normal / Cyst / Stone / Tumor with confidence score
12. Stream A — Grad-CAM saliency map generation; Stream B — Risk stratification
13. Automated diagnostic report generation (PDF: prediction, confidence, Grad-CAM, risk level)
14. AI-powered clinical assistant: condition insight, recommended actions, disclaimer

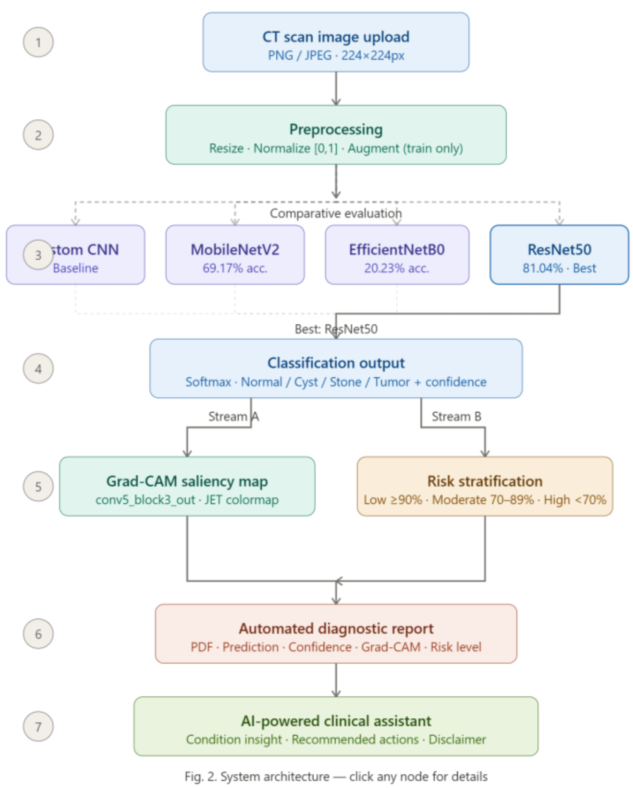


Figure 2: System Architecture / Research Workflow

3.6 Model Architectures

Four architectures were implemented and evaluated under identical hyperparameter and training conditions to ensure a fair comparison.

3.6.1 Custom CNN (Baseline)

The baseline model was built from scratch without any pretrained weights. It comprises three convolutional blocks (Conv2D + BatchNormalization + MaxPooling), followed by Global Average Pooling and two fully connected dense layers (512 and 128 units) with a dropout rate of 0.5. The output layer uses a four-class softmax activation. This model establishes a performance baseline against which transfer learning approaches are benchmarked.

3.6.2 MobileNetV2

MobileNetV2 is a lightweight, computationally efficient architecture designed for mobile and embedded applications. It uses inverted residual blocks with linear bottlenecks, achieving strong performance with significantly fewer parameters than deeper architectures. For this study, ImageNet-pretrained MobileNetV2 weights were used. The original classification head was replaced with a 128-unit dense layer and a dropout of 0.3. Fine-tuning was performed from layer 100 onward after initial frozen training.

3.6.3 EfficientNetB0

EfficientNetB0 achieves performance improvements by systematically scaling network depth, width, and input resolution using a compound scaling coefficient. It employs MBConv blocks with squeeze-and-excitation modules. Despite its strong ImageNet performance, EfficientNetB0 exhibited severe difficulties adapting its RGB-optimized pretrained features to grayscale CT scan data, resulting in near-random classification performance (20.23% accuracy).

3.6.4 ResNet50

ResNet50 employs residual (skip) connections that allow gradient flow across 48 convolutional layers, effectively mitigating the vanishing gradient problem that limits the depth of traditional CNN architectures. With approximately 25.6 million parameters organized into four residual stages, ResNet50 can learn complex multi-scale features — including subtle differences in tissue texture, density, and boundary characteristics — that are critical for kidney lesion differentiation. A custom classification head was added: Global Average Pooling, a 256-unit dense layer, dropout of 0.3, and a four-class softmax output. ResNet50 achieved the best performance across all evaluation metrics.

Hyperparameter	Custom CNN	MobileNetV2	EfficientNetB0	ResNet50
Optimizer	Adam	Adam	Adam	Adam
LR Phase 1	0.001	0.001	0.001	0.001
LR Phase 2	N/A	0.0001	0.0001	0.0001
Batch Size	32	32	32	32
Epochs (Ph. 1)	25	10	10	10
Epochs (Ph. 2)	N/A	20	20	20
Dropout	0.5	0.3	0.3	0.3
Dense Units	512, 128	128	128	256

Table 3: Hyperparameter Configuration for All Four Models

3.7 Transfer Learning Strategy

Each of the three pretrained models (MobileNetV2, EfficientNetB0, ResNet50) was trained in two phases to effectively leverage ImageNet knowledge while adapting to CT scan data:

Phase 1 — Feature Extraction: The backbone was completely frozen. Only the new classification head was trained for 10 epochs using a learning rate of 0.001. This allowed the new layers to learn meaningful representations without disturbing the pretrained backbone features.

Phase 2 — Fine-Tuning: Selected deeper backbone layers were unfrozen and the entire network was trained jointly for an additional 20 epochs using a reduced learning rate of 0.0001. This conservative learning rate prevented catastrophic forgetting of ImageNet-derived features while enabling domain-specific adaptation.

3.8 Grad-CAM Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) was integrated into the system to generate visual explanations for model predictions. Grad-CAM computes the gradient of the predicted class score with respect to the feature maps of the final convolutional layer, then applies a weighted combination to produce a spatial heatmap indicating the image regions most influential to the prediction.

For ResNet50, Grad-CAM was applied to the 'conv5_block3_out' layer. The resulting heatmap is upsampled to the original image dimensions and overlaid using a JET colormap, where warm colors (red, yellow) highlight highly influential regions and cool colors (blue) indicate low influence. Inspection of Grad-CAM outputs confirmed that ResNet50 focuses on clinically appropriate anatomical regions — the collecting system for stones and cysts, and the cortex for tumors — consistent with radiological practice.

3.9 Confidence-Aware Risk Stratification

The softmax output layer produces a probability score for each of the four classes. The maximum probability score is taken as the prediction confidence. Rather than presenting this raw score directly, a three-level risk stratification framework was developed to translate model confidence into clinically actionable guidance:

Confidence	Risk Level	Clinical Action
≥ 90%	Low Risk	Proceed; routine expert review advised
70–89%	Moderate Risk	Expert review is strongly recommended before taking action
< 70%	High / Uncertain	Expert review required; decisions should not be based solely on model output

Table 4: Risk Level Framework Based on Confidence Score

3.10 AI Assistant and Diagnostic Reports

An AI-powered clinical assistant was integrated into the system to provide users with condition-specific information following a prediction. The assistant communicates possible symptoms to monitor, suggested next steps based on clinical guidelines, and basic lifestyle recommendations. It includes explicit disclaimers clarifying that the system is a support tool and does not constitute a clinical diagnosis.

The system also automatically generates a two-page PDF diagnostic report after each prediction. The report includes: the predicted condition, model confidence score, risk level, Grad-CAM visualization, class probability distribution for all four classes, recommended next steps, and a disclaimer regarding the system's limitations and intended use.

3.11 Evaluation Metrics

Model performance was evaluated on the held-out validation set using macro-averaged metrics, ensuring equal weight is given to each class regardless of sample size — a critical consideration given the class imbalance in the dataset.

Metric	Formula	Clinical Interpretation
Accuracy	(TP + TN) / Total	Overall correct predictions; can be misleading on imbalanced datasets
Precision	TP / (TP + FP)	Of all predicted positives, proportion truly positive; minimises false alarms
Recall (Sensitivity)	TP / (TP + FN)	Of all true positives, proportion correctly identified; critical for clinical screening
F1-Score	2 × (P × R) / (P + R)	Harmonic mean of Precision and Recall; balanced metric for imbalanced classes
Confusion Matrix	Per-class prediction breakdown	Detailed error analysis across all classification outcomes

Table 5: Evaluation Metrics Definitions, Formulas, and Clinical Interpretation

3.12 Implementation Tools

The following tools and libraries were used for implementation:

- Programming Language: Python 3.9
- Deep Learning Framework: TensorFlow 2.x / Keras
- Web Interface: Streamlit
- Data Manipulation: NumPy, Pandas
- Image Processing: OpenCV, PIL (Pillow)
- Visualization: Matplotlib, Seaborn
- Report Generation: ReportLab (PDF generation)
- Model Persistence: TensorFlow SavedModel format
- Development Environment: Google Colab / Jupyter Notebook

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results of model training, evaluation, and clinical interface testing. Performance was assessed on a validation set of 1,149 images using macro-averaged metrics.

4.1 Model Performance Comparison

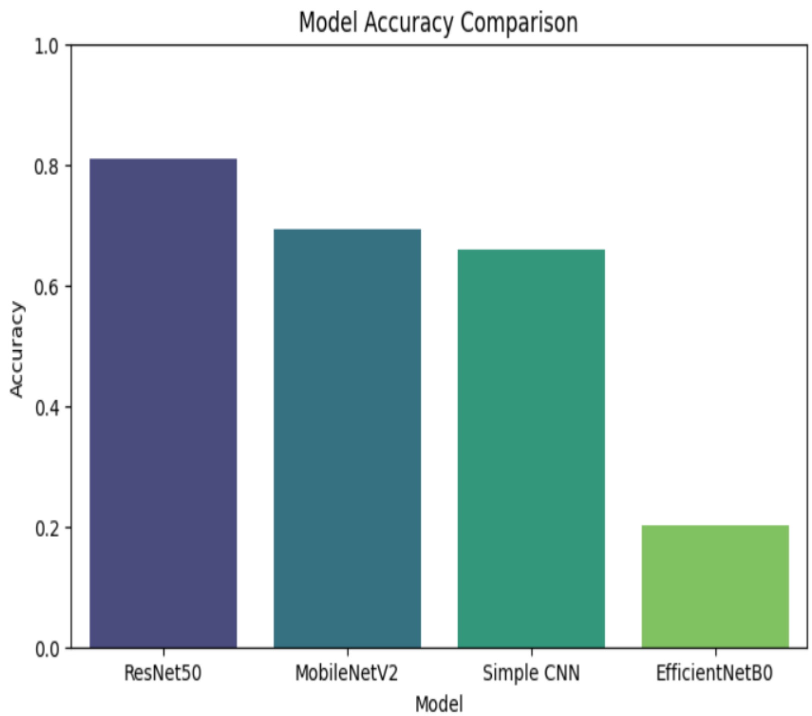


Figure 12: Model Accuracy Comparison Chart

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN (Baseline)	65.99%	0.671	0.660	0.663
MobileNetV2	69.17%	0.879	0.692	0.762
EfficientNetB0	20.23%	0.058	0.202	0.089
ResNet50 (Best)	81.04%	0.865	0.810	0.819

Table 6: Validation Performance of All Four Architectures (Macro-Averaged)

ResNet50 outperformed all other architectures across every evaluation metric. Its residual connections enable the network to learn deep, multi-scale feature representations without suffering from vanishing gradients — a key advantage when distinguishing subtle differences in kidney lesion texture, density, and morphology.

The Custom CNN baseline achieved 65.99% accuracy, confirming the substantial performance advantage offered by ImageNet pretraining even when only the final classification layers are adapted. MobileNetV2 demonstrated high precision (0.879) but comparatively low recall (0.692), indicating that while its positive predictions were mostly correct, it missed a significant proportion of true lesion cases — a clinically concerning pattern in a screening context.

EfficientNetB0's near-random performance (20.23%) was the most striking finding. This model is designed for RGB natural images and its pretrained features proved unable to transfer effectively to grayscale CT scan data. This finding highlights an important principle: ImageNet performance does not guarantee performance on medical imaging tasks, and domain-specific validation is essential before deploying any pretrained model in a clinical context.

4.2 Confusion Matrix Analysis

Confusion matrices were generated for all four models to provide a detailed breakdown of classification behaviour across all lesion types.

4.2.1 ResNet50

ResNet50 correctly classified 488 Cyst, 570 Normal, 139 Stone, and 333 Tumor cases. The most notable error was the misclassification of 192 Normal cases as Tumor. While not ideal, this type of error — flagging a healthy case for specialist review — is clinically preferable to the converse (missing an actual tumor). The Stone class showed the lowest recall (65.3%), likely attributable to the limited Stone training data and the visual similarity between small stones and blood vessels on CT scans.

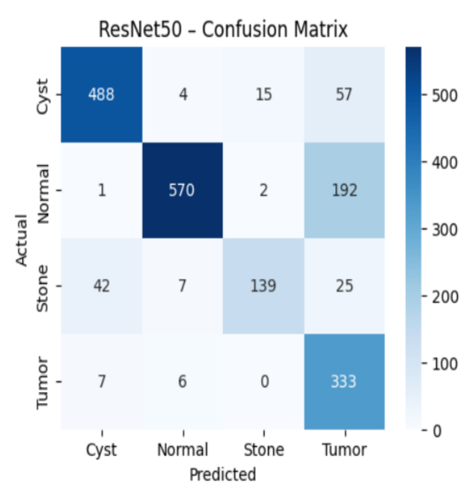


Figure 3: ResNet50 Confusion Matrix

4.2.2 MobileNetV2

MobileNetV2 achieved perfect Stone class recall — correctly identifying all 213 Stone cases in the validation set. However, it catastrophically misclassified 397 out of 765 Normal cases as Stone, which would be clinically unacceptable as it would expose healthy patients to unnecessary intervention. This pattern of error illustrates the risk of evaluating models solely on aggregate accuracy metrics without examining per-class error distributions.

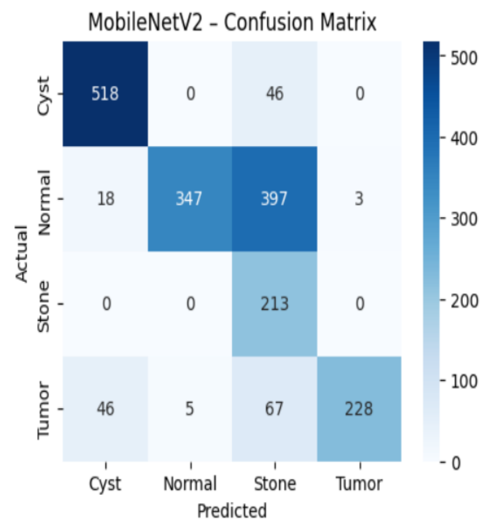


Figure 4: MobileNetV2 Confusion Matrix

4.2.3 Custom CNN

The Custom CNN's confusion matrix reflects the expected limitations of a model trained from scratch on a limited dataset. Classification errors are distributed across all classes, with no dominant error pattern. The model lacks the depth and feature richness provided by pretrained ImageNet weights, resulting in comparatively poor discrimination between visually similar conditions.

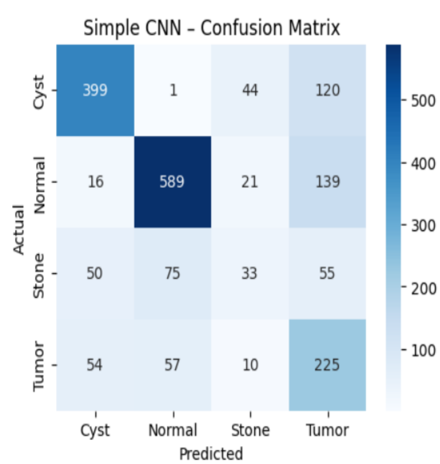


Figure 5: Custom CNN Confusion Matrix

4.2.4 EfficientNetB0

EfficientNetB0's confusion matrix reveals that the model collapsed to predicting primarily one or two classes for all inputs, indicating a complete failure to learn discriminative patterns from the CT scan data. This behaviour is consistent with a domain transfer failure — the model's RGB-optimized feature extractors produced representations that were not meaningful for grayscale medical images.

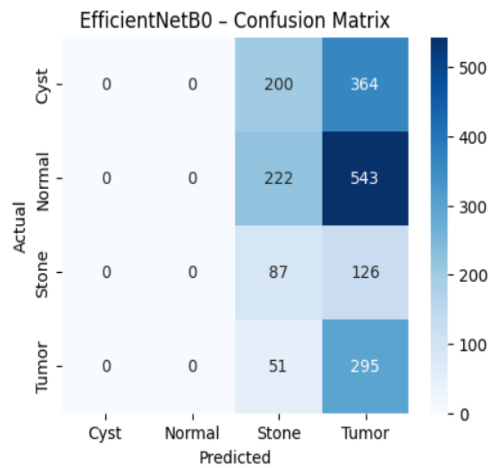


Figure 6: EfficientNetB0 Confusion Matrix

4.3 Grad-CAM and Clinical Interface

Grad-CAM analysis was applied to ResNet50 predictions across all four classes. Qualitative inspection of the resulting heatmaps confirmed that the model consistently directs attention to clinically appropriate anatomical regions:

- Stone predictions: The model highlights the renal collecting system — the anatomical region where stones typically form and lodge.
- Cyst predictions: Attention is directed to fluid-filled spaces in the kidney parenchyma, consistent with cyst location.
- Tumor predictions: The model focuses on the renal cortex and areas of irregular tissue density, consistent with radiological tumor assessment criteria.
- Normal predictions: Attention is more diffuse, with no specific focal abnormality highlighted.

This alignment between model attention and clinical knowledge is a strong indicator that ResNet50 is learning genuinely useful medical features rather than spurious correlations with image background or scanner artefacts.

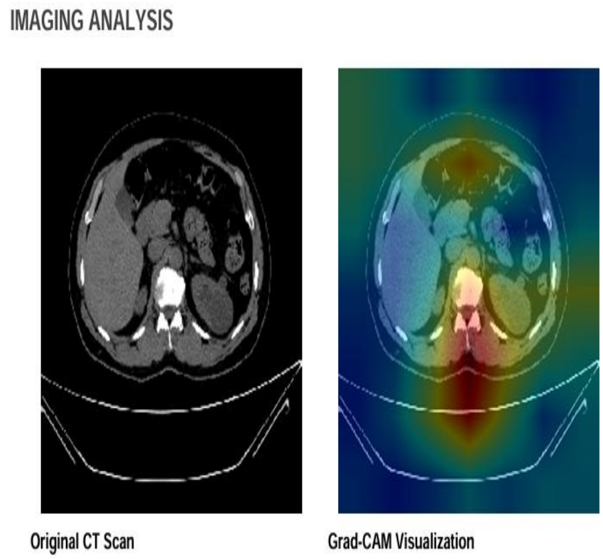


Figure 7: Grad-CAM Visualization for a Kidney Stone Prediction

Figure 7 illustrates the Grad-CAM output for a Stone prediction. The warm-colored regions correspond precisely to the area of the kidney where stones are clinically expected — validating that the model's high-confidence prediction is grounded in anatomically appropriate evidence.

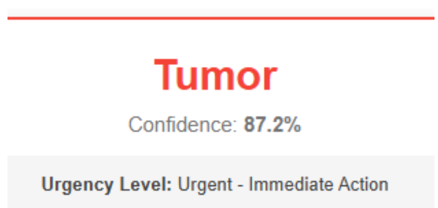
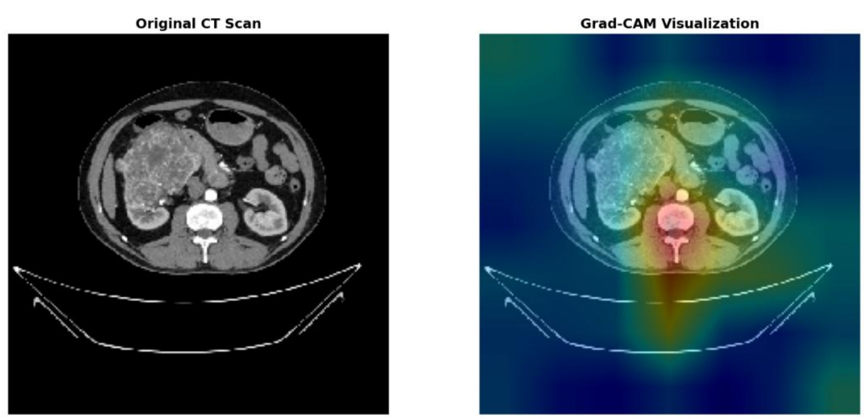


Figure 8: Uploaded CT Scan Interface



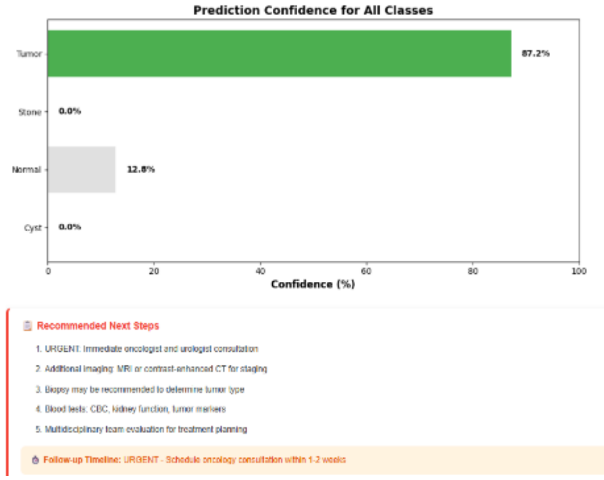
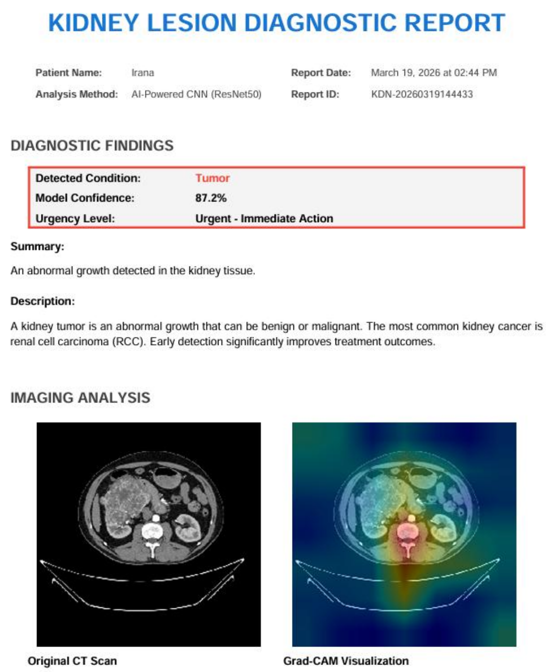


Figure 9: Predicted Class and Confidence Output



KIDNEY LESION DIAGNOSTIC REPORT

Patient Name:	Irana	**Report Date:**	March 19, 2026 at 02:44 PM
Analysis Method:	AI-Powered CNN (ResNet50)	**Report ID:**	KDN-20260319144433

DIAGNOSTIC FINDINGS

Detected Condition:	Tumor
Model Confidence:	**87.2%**
Urgency Level:	**Urgent - Immediate Action**

Summary:

An abnormal growth detected in the kidney tissue.

Description:

A kidney tumor is an abnormal growth that can be benign or malignant. The most common kidney cancer is renal cell carcinoma (RCC). Early detection significantly improves treatment outcomes.

IMAGING ANALYSIS

Original CT Scan **Grad-CAM Visualization**

Figure 10: Report

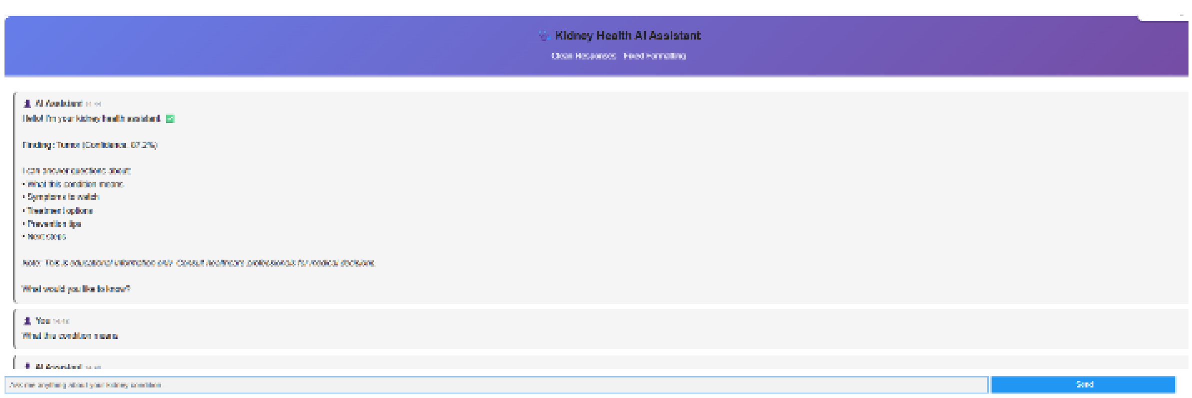


Figure 11: AI Clinical Assistant Output

4.4 Comparison with Existing Models

The proposed system is compared against relevant existing approaches in the literature:

Reference	Model	Accuracy	Dataset	Explainability
Subedi et al.	Vision Transformer	99.64%	CT Kidney (Kaggle)	No
Pimpalkar et al.	InceptionV3	99.96%	CT Kidney (Kaggle)	No
Alzu'bi et al.	2D-CNN	97.70%	Custom (KAUH)	No
Mehedi et al.	VGG16 + Seg.	99.48%	CT Kidney (Kaggle)	No
Ahmed et al.	VGG16 + Grad-CAM	97.41%	KUB X-ray	Yes
Proposed System	ResNet50 + Grad-CAM + AI Assistant	81.04%	CT Kidney (Kaggle)	Yes (Full Suite)

Table 8: Comparison with Existing Models in Literature

While the proposed system's accuracy (81.04%) is lower than several cited studies, this figure reflects a more conservative and methodologically rigorous evaluation approach. Many high-accuracy studies used the same Kaggle dataset without external validation, and some may have benefited from favourable train-test splits that allowed data leakage. A model achieving 99%+ accuracy on a single dataset without independent validation does not reliably predict performance in real clinical environments with different scanners, imaging protocols, or patient demographics.

More importantly, the proposed system delivers capabilities that no other reviewed study provides as a unified pipeline: Grad-CAM explainability, confidence-aware risk stratification, automated PDF report generation, and an AI clinical assistant. These features collectively address the clinical utility gap that accuracy-focused research has failed to bridge.

4.5 Discussion of Findings

4.5.1 Why ResNet50 Performed Best

ResNet50's superior performance can be attributed to its architectural depth and skip connection design. The residual connections allow the network to learn identity mappings as a fallback, enabling very deep networks (50 layers) to train stably without gradient degradation. This depth allows the model to extract features at multiple levels of abstraction — from low-level edge and texture features in shallow layers to complex structural patterns in deeper layers. These multi-scale features are essential for differentiating kidney lesions that can appear subtly different in texture, density, and boundary characteristics on CT scans. ResNet50's precision of 0.865 is particularly valuable in a clinical context, as it minimizes the rate of false positive diagnoses that could lead to unnecessary patient anxiety and intervention.

4.5.2 Why EfficientNetB0 Failed

EfficientNetB0's catastrophic failure on this task provides a valuable cautionary lesson. The model was pretrained on RGB natural images from ImageNet and its compound scaling strategy was optimized for that domain. CT scan images are fundamentally different: they are grayscale (or pseudo-colored), they encode tissue density through Hounsfield units, and their visual patterns bear little resemblance to natural photographs. EfficientNetB0's pretrained feature extractors — optimized for color-based texture and object recognition — produced representations that were not meaningful for CT scan content, preventing the model from learning discriminative patterns during fine-tuning. This finding reinforces the critical importance of domain-specific model evaluation before clinical deployment.

4.5.3 Clinical Positioning and Responsible AI

The system is explicitly designed as a clinical decision support tool, not a replacement for physician judgment. The Grad-CAM visualizations empower clinicians to verify model reasoning, the risk stratification framework communicates uncertainty in actionable terms, and the AI assistant provides informational context rather than diagnostic conclusions. All system outputs include explicit disclaimers clarifying the system's limitations.

This design philosophy reflects a responsible AI approach that prioritizes transparency, human oversight, and patient safety alongside technical performance. The system is particularly well-suited to supporting general practitioners and healthcare workers in resource-limited settings who need initial screening guidance before specialist review is available.

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN (Baseline)	65.99%	0.671	0.660	0.663
MobileNetV2	69.17%	0.879	0.692	0.762
EfficientNetB0	20.23%	0.058	0.202	0.089
ResNet50 (Best)	81.04%	0.865	0.810	0.819

Table 9: Comprehensive Comparative Summary – All Models

CHAPTER 5

FUTURE PROSPECTS

5.1 Limitations of the Current Study

While the system demonstrates strong performance and clinical utility, several limitations must be acknowledged:

Dataset Scope: The system was trained and evaluated exclusively on the CT Kidney Dataset from Kaggle. While this dataset is widely used in the research community, it was collected from a limited number of imaging centers and scanners. Model performance on CT scans acquired with different imaging protocols, scanner manufacturers, or patient demographics may differ, and generalizability cannot be guaranteed without external validation.

Single-Image Analysis: The system processes individual CT images in isolation and does not consider 3D volumetric information, multi-slice context, or patient clinical metadata (e.g., age, medical history, laboratory results). Clinical diagnosis typically integrates multiple sources of information, and the absence of these inputs limits the system's diagnostic depth.

Lesion Morphology: The system classifies lesions but does not quantify lesion size, shape, or precise location — information that is clinically important for treatment planning and monitoring disease progression over time.

Stone Class Performance: The Stone class achieved the lowest recall (65.3%) among the four classes, likely due to the combination of limited training examples and the visual similarity of small stones to blood vessels on CT scans. Improved performance on this class requires either more training data or targeted augmentation strategies.

AI Assistant Generalization: The AI clinical assistant provides condition-level guidance that is consistent across all patients with the same predicted condition. It does not personalize recommendations based on individual patient characteristics, medical history, or comorbidities.

Computational Requirements: ResNet50 with Grad-CAM computation requires meaningful computational resources. While the system was deployed via Streamlit on cloud infrastructure for this study, deployment in low-resource clinical environments may require model compression or lightweight alternatives.

5.2 Future Research Directions

Several meaningful directions exist for extending and improving the system:

- External Clinical Validation: The most critical next step is validating the system on independent clinical datasets from different hospitals and scanner types to assess real-world generalizability. Collaboration with radiology departments to collect annotated datasets would significantly strengthen this work.
- 3D Volumetric Analysis: Extending the system to process full 3D CT volumes rather than individual 2D slices would provide richer spatial context and improve diagnostic accuracy — particularly for detecting small stones and delineating tumor boundaries.
- Multi-Modal Data Integration: Incorporating patient clinical metadata (age, symptoms, laboratory results, medical history) alongside CT image analysis could substantially improve prediction accuracy and enable personalized risk stratification.
- Lesion Segmentation: Adding a segmentation module (e.g., using UNet or Mask R-CNN) to delineate lesion boundaries and quantify lesion size would provide clinically essential information for treatment planning.
- Ensemble Methods: Combining ResNet50 and MobileNetV2 in an ensemble could leverage their complementary strengths — ResNet50's overall accuracy and MobileNetV2's Stone detection — to improve per-class recall while maintaining precision.
- Explainability Enhancements: Supplementing Grad-CAM with SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) would provide feature-level attribution alongside spatial heatmaps, further strengthening clinical interpretability.
- Federated Learning: Training the model across multiple hospitals using federated learning would enable access to diverse datasets without compromising patient data privacy — a critical consideration for clinical AI deployment.

- Longitudinal Monitoring: Extending the system to support sequential analysis of CT images taken at different time points would enable monitoring of lesion progression or response to treatment over time.

5.3 Practical Implementation and Deployment

A key future prospect is the integration of this system into clinical workflow as a validated clinical decision support system (CDSS). Such a system would be embedded within hospital radiology information systems (RIS) or picture archiving and communication systems (PACS), enabling radiologists to receive automated pre-screening results alongside the raw CT images.

For practical deployment, the system requires prospective clinical validation on real patient data under institutional ethics approval, regulatory compliance review (e.g., FDA clearance or CE marking depending on jurisdiction), and ongoing monitoring for performance drift as patient populations and imaging technologies evolve.

The Streamlit-based web interface developed in this study provides a proof-of-concept deployment platform that could be extended into a standalone mobile or web application for use in primary care or community health settings — extending access to AI-assisted kidney lesion screening beyond major specialist centers.

5.4 Conclusion

This study presents the design, implementation, and evaluation of a comprehensive Automated Kidney Lesion Classification System using CNNs with transfer learning. The system classifies CT images into four clinically meaningful categories — Normal, Cyst, Stone, and Tumor — and integrates explainability, risk stratification, automated reporting, and AI-assisted consultation into a unified clinical support pipeline.

Among the four architectures evaluated, ResNet50 achieved the best performance with an accuracy of 81.04%, a precision of 0.865, a recall of 0.810, and an F1-score of 0.819. EfficientNetB0's near-random performance (20.23%) demonstrated the critical importance of domain-specific model evaluation — ImageNet performance does not guarantee transferability to medical imaging tasks.

Grad-CAM analysis confirmed that ResNet50 focuses on clinically appropriate anatomical regions for each predicted class, providing evidence that the model's predictions are grounded in medically meaningful features rather than spurious correlations. The confidence-aware risk stratification framework and AI clinical assistant extend the system's utility beyond classification alone, supporting more nuanced and actionable clinical guidance.

What distinguishes this system from the majority of existing work is not its classification accuracy in isolation, but rather its integration of accuracy, explainability, uncertainty quantification, automated reporting, and clinical assistance into a single, coherent, and practically deployable pipeline. As a proof-of-concept, the system demonstrates a clear and principled approach to building AI diagnostic tools that are not only technically capable but also transparent, trustworthy, and clinically useful.

Future work should focus on external clinical validation, 3D volumetric analysis, multi-modal data integration, and federated learning to further strengthen the system's performance and generalizability. With these enhancements, the system has the potential to meaningfully support early kidney lesion detection and improve patient outcomes across diverse healthcare settings.

CHAPTER 6

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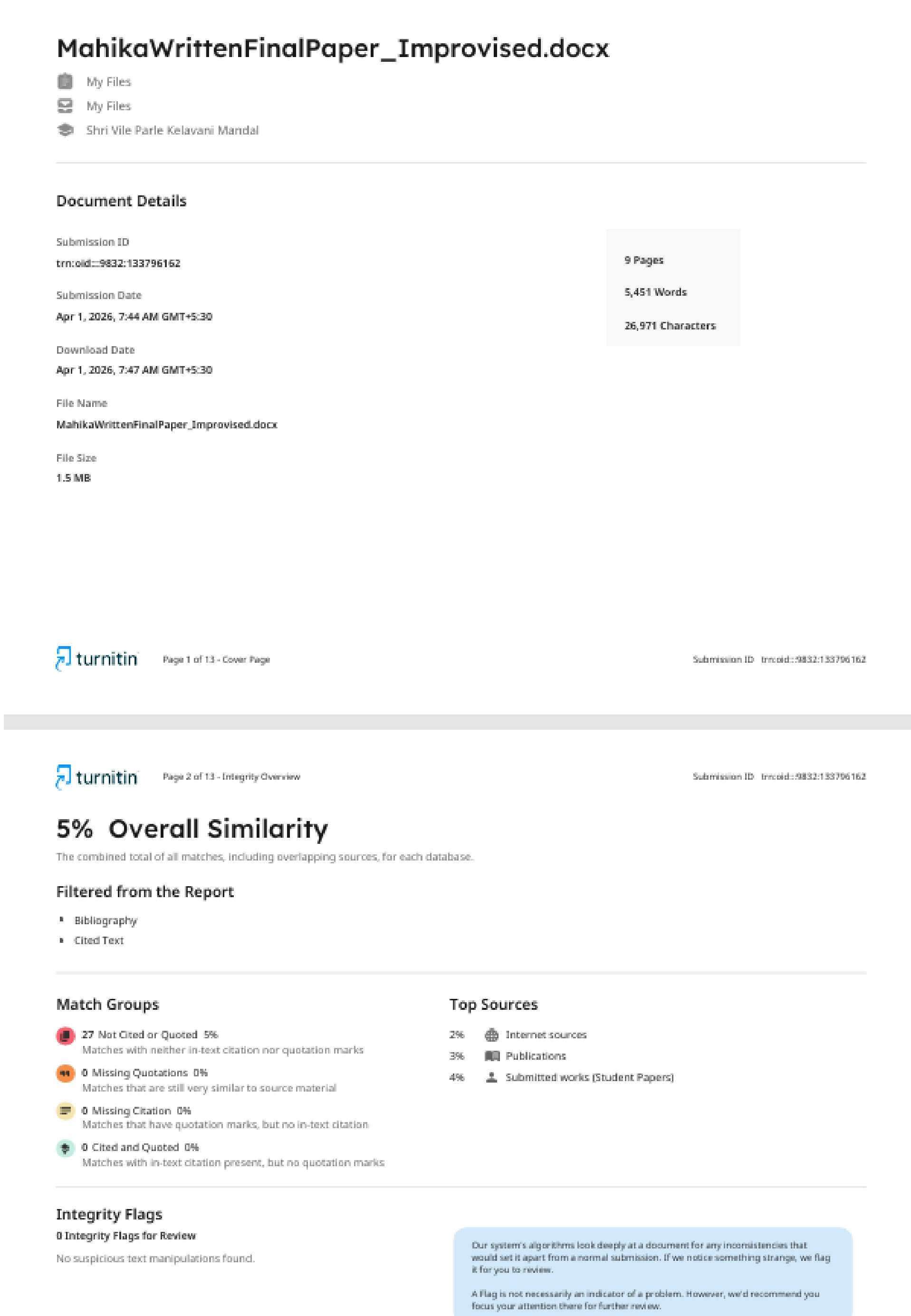
CHAPTER 7

PLAGARISM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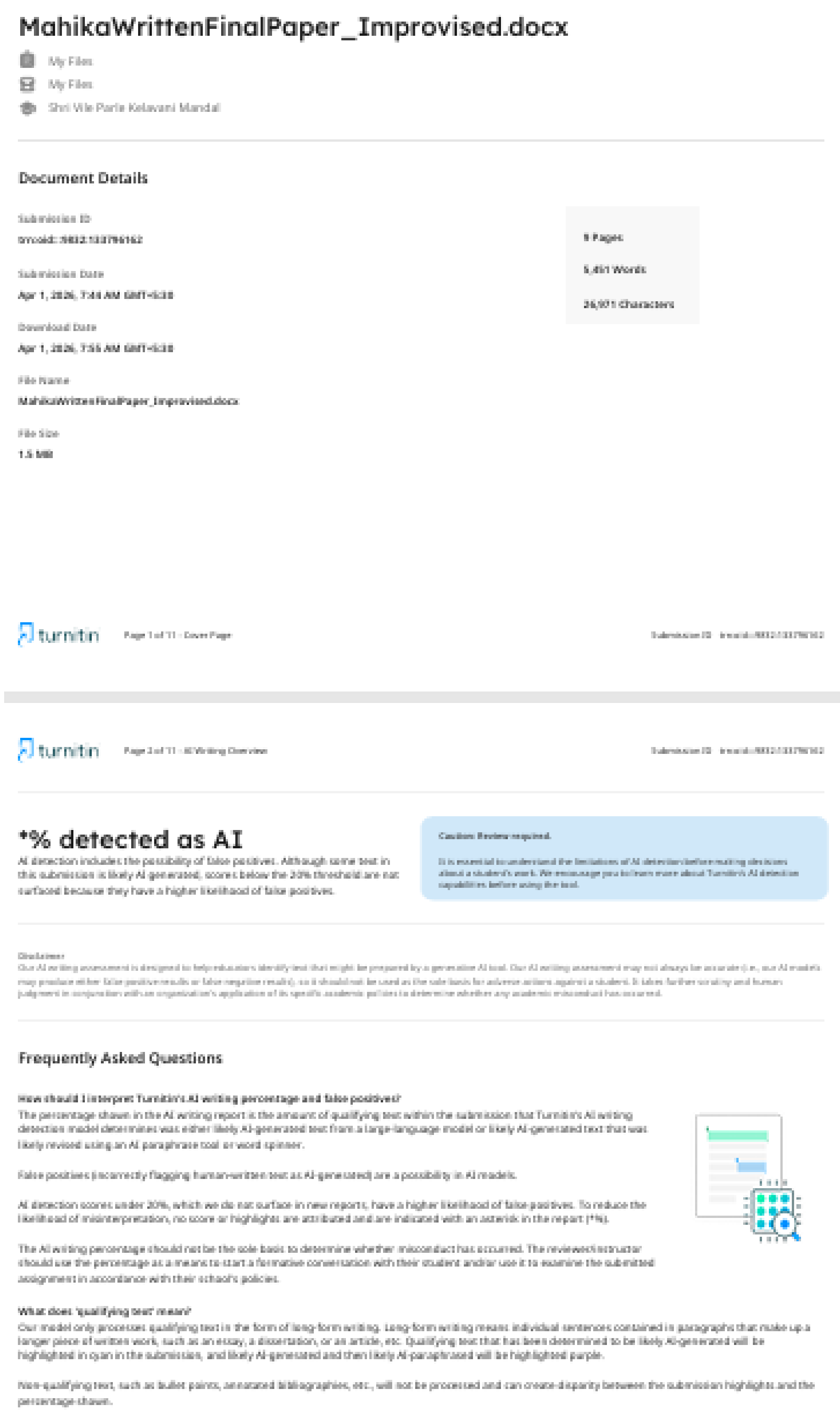

Fig. 14 AI detection report

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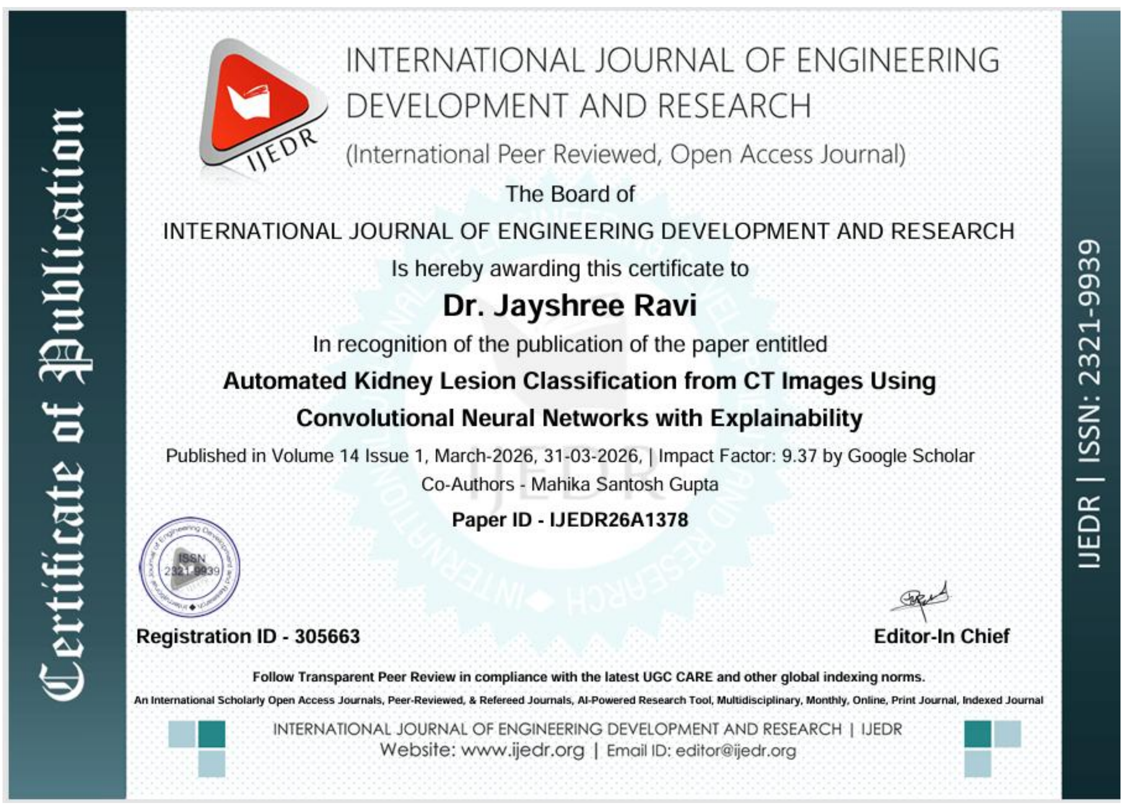

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Automated Kidney Lesion Classification from CT Images Using Convolutional Neural Networks with Explainability

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ABSTRACT

Nowadays, due to people's lifestyle habits such as unhealthy diets, less exercise, and stress are leading to more kidney-related issues. There are three types of kidney lesions that can be seen on CT scans: cysts, stones and tumors. These conditions can have different outcomes depending on how early they are diagnosed. Looking at CT scans to find these problems is a lot of work. Most importantly it depends on the person looking at the scans. This is especially true in places where they do not have a lot of resources.

This article is about Automated Kidney Lesion Classification from CT Images which can automatically classify kidney lesions from CT images using Convolutional Neural Networks to tell if the kidney is normal or if there is a cyst, stone or tumor. What it does exactly is it takes your uploaded CT scanned image of kidney then processes it and gives you the analysis of it with the results. It makes a map that highlights the problem areas with the help of Grad-CAM. It also shows the accuracy using confidence bar. Finally generates a detailed report of the patient and also aims to answer any queries with the help of AI chat assistant.

Keywords: *Kidney Lesion Classification, CNN, Transfer Learning, Grad-CAM, Explainable AI, CT Imaging, ResNet50, Clinical Decision Support*

PROBLEM STATEMENT

Automated kidney lesion classification has seen a lot of progress over the past few years. However, there are still many issues which may result in serious problems.

One of the problems is that kidney lesions are often very small and hard to see in CT scans. Hence, it makes tough for doctors to treat them correctly. Kidney lesions can also look a lot alike which makes it hard to tell them apart.

There is also no explanation of how kidney lesion classification systems work. In places where resources are limited it is even harder to manage a lot of patients and get it right. If we make a mistake with kidney lesion classification it is not a technical issue. This means that the patient will not get the treatment, for their kidney lesion.

This project is trying to solve these kidney lesion classification problems.

I.INTRODUCTION:

Kidney disease is a health problem worldwide affecting over about 850 million peopleThe three main types of kidney lesions that can be seen on CT scans are cysts, stones and tumors. These conditions can have different outcomes depending on how early they are diagnosed. For example if kidney cancer is diagnosed early the five-year survival rate is over 93%. However if it is not diagnosed until later the survival rate drops to below 12%. This highlights the importance of timely diagnosis.

CT scans are the way that kidney lesions are diagnosed. However interpreting CT scans is an time-consuming task that requires a lot of expertise. Primarily radiologists need to have an in depth knowledge about various conditions in addition they need to make the decision on basis of what they see.

This can be challenging in cases where the lesions are small or not clearly visible. In healthcare settings with the increasing number of patients it becomes challenging to keep up the demands.

The use of deep learning has increased in past few years for analyzing medical images. Convolutional Neural Networks (CNNs) are a type of deep learning model which can be used to classify images. These CNNs work well as the look at pictures and figure out what is in them. They do this by checking out the dots of color in a picture called pixels and using that to guess what the picture shows to make predictions.

The main goal of this project was to create a system that can be used to classify kidney lesions from CT images using Convolutional Neural Networks.

This system is designed to be easy to understand so it can tell us how it makes its predictions.

This system takes care of everything from uploading the image to giving us the results. Here is what it does:

- Compares four models with different architectures to see which one is the best.
- Grad-cam to explain every prediction it makes
- It helps figure out the level of risk based on how confident it is with the prediction.
- It generates reports and also provides more knowledge about the condition with AI chat assistant.

II.LITERATURE REVIEW

Nowadays there has a been a significant rise of AI based models in healthcare sector. Deep learning for kidney lesion classification has become more popular over the few years.It's because of their potential to improve the precision of and help with faster more reliable results for patients.
There have been several studies on kidney lesion classification using deep learning.

Subedi et al [5] compared ResNet50, VGG16 and a Vision Transformer on the same Kaggle CT dataset.
The Vision Transformer got a high accuracy of 99.64% on a 90:10 split.However, it does not explain how it makes theses decisions and lacks support to back it.

Pande and Agarwal [6] used YOLOv8n-cls. and got an accuracy of 82.52%. YOLO is fast but it does not extract features in depth.
Mehedi and others [7] showed that adding segmentation with UNet and VGG16 increases accuracy to 99.48%.
This method requires pixel-level annotation but Clinical datasets do not have this level of detail.

In transfer learning Pimpalkar et al.[10] Got an accuracy of 99.96%, with a changed InceptionV3 model and data augmentation.However,it can be difficult to get this result if you do not have the same setup. The model might only work well with their specific dataset and not on different datsets. Pimpalkar et al. used transfer learning approaches to get this result.

Majid et al. [11] worked with DenseNet-121 and got great results achieving 98.22% accuracy. Along with that, Ahmed et al. [12] used a different approach by using VGG16 model with Grad-CAM on X-ray images and achieved 97.41% accuracy.
These works are important because it helps us understand why the DenseNet-121 and VGG16 models make the decisions they do.
The DenseNet-121 and VGG16 models are really good, at what they do and understanding how they work is crucial.
Despite having high accuracy there is a big gap, in current research.

Most systems focus on accuracy. Ignore other key aspects. These aspects include explainability, uncertainty and clinical support.
Many studies also do not validate their results on datasets.
We need to be more careful when we see high accuracy on a particular dataset. Keeping this in mind, our system is not just about performance numbers.

The kidney lesion classification system is designed to enhance knowledge of people and also doctors who are still new in the field to help them get wider picture of the condition.
It is made to help them. It tries to give them a more complete and practical solution, for kidney lesion classification.

III.OBJECTIVES

1. Build a system that looks at pictures of kidneys and figures out if the kidneys are normal or if they have something with them like a cyst or a stone or a tumor.
2. Compare four different models which are, a custom model, MobileNetV2, EfficientNetB0 and ResNet50 to see which one works best for our dataset.
3. Use Grad-CAM to create pictures that show doctors which parts of the kidney image are affected and the computer focused on to make its decision.
4. Create a dashboard that tells doctors how sure the computer is about its decision using the confidence bars.
5. Build a system that helps doctors and patients with a treatment plan based on the illness they are dealing with.
6. Check if the system actually works well by looking at curacy, precision, recall, F1-score and confusion matrix analysis.

IV.CONTRIBUTIONS

1. We compared four different models to see which one works best with the same dataset.
2. We used Grad-CAM to highlight the areas on a scan that a doctor would look at so we know the system is making decisions based on medical evidence.
3. We sort patients by risk level based on how sure we are using the confidence score.
4. We built an AI assistant which can be helpful for doctors decide what to do based on what the system predicts and also help the patients to learn more about their condition.
5. We created a way to download the report of the prediction that is easy for patients and doctors to use and understand.

V.METHODOLOGY

A. Dataset

The project uses a CT Kidney Dataset that was first shared on Kaggle by Islam et al. [16]. This CT Kidney Dataset originally had 12,446 images. However, the dataset for this project used a smaller set with a better balance of images.
The CT Kidney Dataset used for the project has 5,708 CT images that are divided into four groups:
Cyst: 1,500 images
Normal: 1,500 images
Tumor: 1,500 images
Stone: 1,208 images

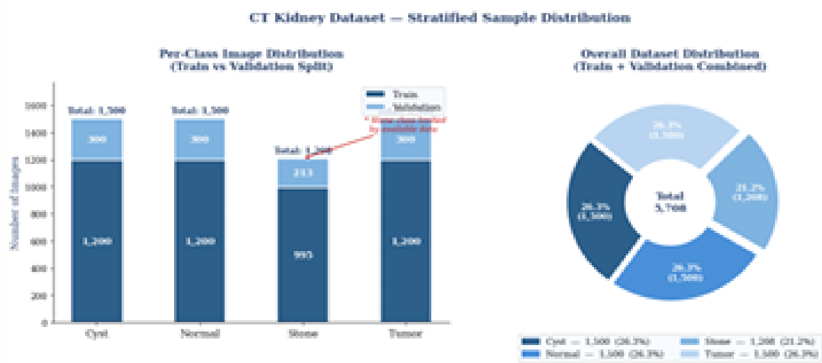


Fig. 1. Distribution of dataset classes (total = 5,708 images). The ***Stone*** *class has the fewest samples (11.1%), so we handled this imbalance using class weights and some extra sampling during training.*

The CT Kidney Dataset was split into two parts: one for training and one for validation. It was ensured that the number of images in each group was similar in both parts. Out of all the images in the CT Kidney Dataset:
4,559 images were used to train the model
1,149 images were used to validate the model

For most of the groups, such as the Cyst group, the Normal group and the Tumor group we used 1,200 images to train the model and 300 images to validate the model. The Stone group had images so the project team used 995 images to train the model and 213 images to validate the model.

The validation part of the CT Kidney Dataset was kept separate, from the training part. We did not change or add to the validation images so the results show how well the model really works with images it has not seen before.

B. Preprocessing and Augmentation

All the images were made to be the size, which is 224 by 224 pixels. The color of the pixels in the images was also made to be the same by dividing the numbers by 255 so they are all between 0 and 1.
Few changes were made on the images to make the model better at recognizing kidneys.

It was done ImageDataGenerator from Keras. The images were rotated a bit up to 15 degrees to make the model see the kidneys from angles. They were also made a bit wider or narrower up to 5 percent to make the model better at recognizing kidneys that're not perfect. Sometimes the image was flipped horizontally like it was mirrored, to make the model see the kidneys from the side.

The images were also made a bit smaller up to 10 percent to make the model better at recognizing kidneys of different sizes.
The empty spaces in the images were filled with the color of the pixel next to it which is the color of the kidney.

The model was not shown any images that were flipped down because those images would not show a kidney.
We only want the model to recognize pictures of kidneys.
There was a little problem with the data because the Stone images were 11.1% of all the images.

More Stone images were added to the data so the model can see more examples of kidneys, with a Stone.
The model was also told to give weight to the types of images that were not as common like the Stone images so it can recognize them better.

Augmentation Parameter	Value / Configuration
Resize	224 × 224 px (bilinear)
Normalization	1/255.0 → [0, 1]
Rotation	±15°
Width / height shift	±5%
Zoom	±10%
Horizontal flip	Enabled
Vertical flip	Disabled (anatomical constraint)
Fill mode	Nearest neighbor

Table I. Training augmentation parameters

VI.SYSTEM ARCHITECTURE

The system is made up of a series of steps that happen one after the other. First a CT scan is uploaded to the system. Then it is cleaned up and prepared for use. After that it is passed through a trained CNN classifier. The CNN classifier is then used for Grad-CAM analysis. This analysis helps the system figure out which parts of the picture are the important by highlighting the important regions.

Based on this analysis confidence score is calculated. The confidence score is used to decide what next steps should be taken. Finally, the patient can download a detailed pdf report. After this if the patient has any queries regarding their condition, they can use the AI assistant to clear their doubts

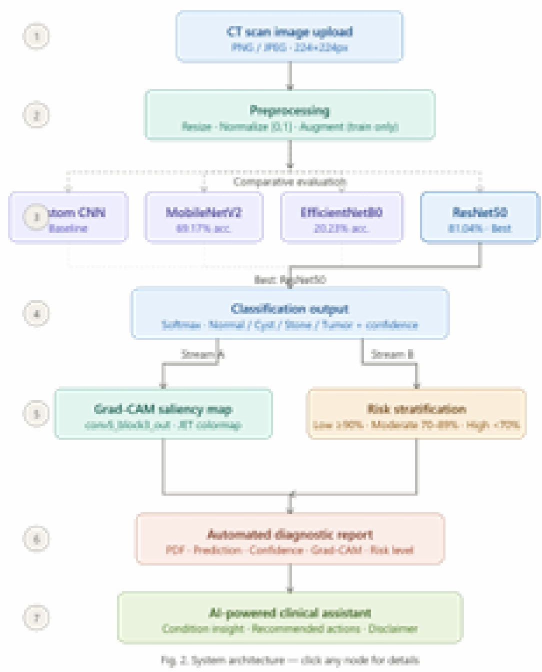


Fig. 2. Flow of the system from start-to-end.

A. Model Architectures

Four different models were built and tested under the same training conditions to make a fair comparison.

1) Custom CNN (Baseline)

The model has three convolutional blocks (Conv2D + BatchNorm + MaxPool), followed by global average pooling. Then it has two dense layers (512 and 128 units), with a dropout of 0.5, and ends with a four-class softmax layer. It was trained from scratch without using any pre-trained weights to establish a baseline performance.

2) MobileNetV2

MobileNetV2 is a good model because it is efficient and uses less computation power. For this project we used MobileNetV2 with weights it already knew from ImageNet. The original final layer was replaced with a dense layer of 128 units and a dropout of 0.3. At first, we did not change most of the layers in MobileNetV2 so the new layers could learn properly. After that, deeper layers (from layer 100 onward) were unfrozen and fine-tuned to improve performance.

3) EfficientNetB0

This model improves performance by scaling depth, width, and image size. It also uses MBConv blocks with squeeze-and-excitation. It also uses the same classification head as MobileNetV2.However the model did not work well when it was used to look at CT scans. The model had trouble when it went from looking at pictures to looking at CT scans as the model was sensitive, to this change from natural images.

4) ResNet50

This model uses residual (skip) connections, which allow it to go deeper (48 layers) without training issues like vanishing gradients. It has around 25.6 million parameters organized into four stages.
The custom classification head includes global average pooling, followed by a 256-unit dense layer, a dropout of 0.3, and a four-class softmax output.
Overall, this model gave the best performance across all evaluation metrics in comparison to other models.

B. Transfer Learning Strategy

Each of the three transfer learning models was trained in two steps. This helped make use of the pre-trained knowledge while adapting to CT scan data.

In Phase 1 the main model, also called the backbone was completely frozen. Only the new classification head was trained for 10 epochs. A small learning rate of 0.001 was used which helps the new layers learn properly without changing the -trained features.

In Phase 2 a few layers, at the end of the model were unfrozen. The whole network was then trained together for another 20 epochs. A smaller learning rate of 0.0001 was used. This smaller learning rate
helped prevent damaging the patterns that the model learned from ImageNet data.

Hyperparameter	Custom CNN	MobileNetV2	EfficientNetB0	ResNet50
Optimizer	Adam	Adam	Adam	Adam
LR Phase 1	0.001	0.001	0.001	0.001
LR Phase 2	N/A	0.0001	0.0001	0.0001
Batch size	32	32	32	32
Epochs (Ph. 1)	25	10	10	10
Epochs (Ph. 2)	N/A	20	20	20
Dropout	0.5	0.3	0.3	0.3
Dense units	512, 128	128	128	256

Table II. Hyperparameter configuration for all four models.

C. Grad-CAM Explainability

Grad-CAM was used to understand how the model makes its predictions. It generates heatmaps that highlight the important areas in the image that influenced the result.

The method looks at the last convolutional layer and calculates how much each feature contributes to a specific class. This contribution is calculated as:

$$\alpha_k^c = \frac{1}{Z}\sum_i\sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

These values are then combined to form the heatmap:

$$\text{Grad-CAM} = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right)$$

The heatmap is then placed over the original image, where warm colors (red, yellow) show important regions and cool colors (blue) show less important ones. Only positive contributions are considered.

For ResNet50, Grad-CAM was applied to the **'conv5_block3_out'** layer.

D. Confidence-Aware Risk Stratification

The model produces a probability score for each of the four respective classes using the softmax outputs. The maximum score was used as the prediction confidence score.

Instead of just showing this number to the user, I created a simple three-level risk grouping system that converts the model's confidence into a clear recommendation on what action to take next.

Confidence	Risk Level	Clinical Action
≥ 90%	**Low Risk**	Proceed; routine expert advised.
70–89%	**Moderate Risk**	expert review is strongly recommended before taking action.
< 70%	**High / Uncertain**	An expert review is required, and decisions should not be made based only on the model output.

Table III. Risk level framework based on confidence score.

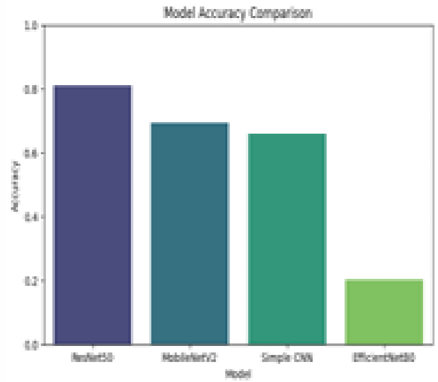
E. AI Assistant and Diagnostic Reports

We have an AI chat assistant through which the patient can clear any doubts that they have. It mentions possible symptoms that should be paid attention to, along with suggesting next steps, it also gives basic lifestyle advice. I want to make it clear that the system is only meant to support people and does not give any diagnosis to them. The advice it gives is based on medical rules.

After the system makes a prediction it automatically creates a two-page PDF report. This report has the condition the system predicted a score that shows how confident it is in its prediction with the level of risk, a Grad-CAM visualization probability for all classes, suggested next steps, and a disclaimer about the system's limitations.

VII.RESULTS

The performance of the models was evaluated using a validation set that had 2,489 images. Macro-averaged metrics were used for this. It means that the performance of the models was evaluated in a way that each class was given the equal importance. The performance of the models was evaluated in this way regardless of how samples each class had. This ensured fair evaluation of the models. It was especially important to evaluate the models in this way because the dataset of images that was used is not balanced.



Model	Accuracy	Precision	Recall	F1-Score
Custom CNN (Baseline)	65.99%	0.671	0.660	0.657
MobileNetV2	69.17%	0.879	0.692	0.718
EfficientNetB0	20.23%	0.058	0.202	0.090
ResNet50 (Best)	**81.04%**	**0.865**	**0.810**	**0.819**

Table IV. Validation performance of all four architectures (macro-averaged).

A.Model Comparison

The ResNet50 model did well compared to all the other models in every way. The ResNet50 model has skip connections which helps it learn more about the pictures making it better to find differences in texture and density between different kidney conditions. The other models were not as good at finding these details. The custom CNN model only got it right 65.99% of the time which shows that using models that have already been trained on lots of pictures, like the ones trained on ImageNet is a good idea even if you only change the last few layers of the ResNet50 model.

MobileNetV2 had a problem. It was really good at being precise and got the predictions right most of the time with a precision of 0.879 but it was not good at finding all the positive cases as its recall was only 0.692. This means that when MobileNetV2 said when it said a case was positive, it was mostly right, but it failed to detect a lot of true cases. In a hospital this can be a big issue because if you miss someone who is really sick it can cause serious harm to patient.

EfficientNetB0 gave the most surprising result with an accuracy of only 20.23%, which is really close to just guessing. This shows that EfficientNetB0 struggled to adapt from images to grayscale CT scans. EfficientNetB0 just could not do it. This highlights a point about EfficientNetB0 and other models that the models which do well on ImageNet may not always do well with medical imaging tasks, like CT scans.

B. Confusion Matrix Analysis

The results for each class from the confusion matrices, which are shown in Figures 3 to 6 give us details about the models. We can learn more about the models beyond looking at the numbers.

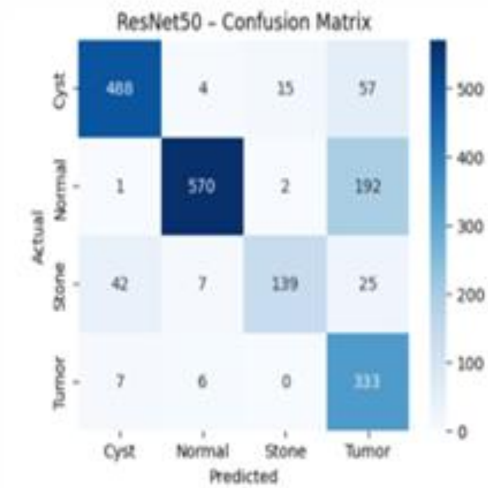


Fig. 3. ResNet50 confusion matrix.

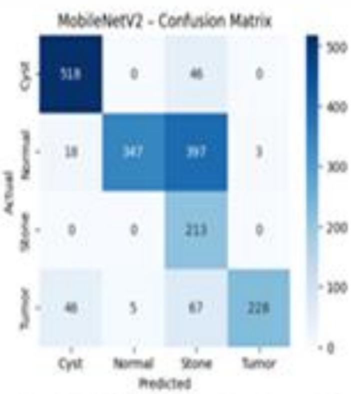


Fig. 4. MobileNetV2 confusion matrix.

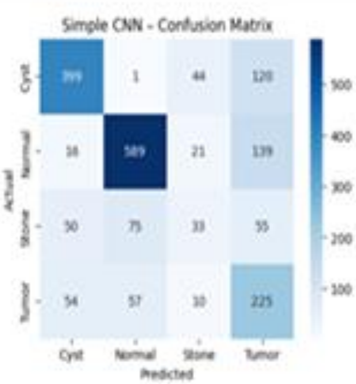


Fig. 5. Custom CNN confusion matrix.

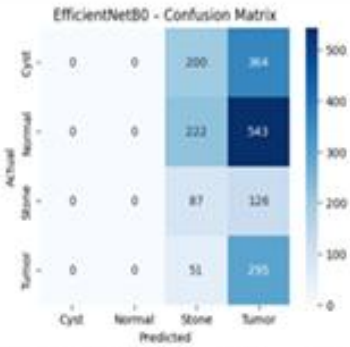


Fig. 6. EfficientNetB0 confusion matrix.

For ResNet50, the model correctly classified 488 Cyst, 570 Normal, 139 Stone, and 333 Tumor cases. The most biggest mistake was that 192 Normal cases were predicted as Tumor. While this is not the most ideal situation, it is still safer in a healthcare settings to flag a healthy case for review than to miss an actual tumor.

The Stone class did the worst with only 65.3% of cases detected. This might be because of the few numbers of stone images in the data and also because small stones look like blood vessels, in CT scans.

MobileNetV2 had a different kind of mistake. It did a good job with the Stone class by correctly classifying all 213 cases. However, it wrongly predicted Normal cases as Stone with just 397 out of 765 which is not safe.

When we look at the results of ResNet50 and MobileNetV2 they both have different strengths and weaknesses. This suggests that combining them (as an ensemble) could help balance their limitations and improve their overall performance.

The Custom CNN did not do well since it was trained from scratch, this makes sense because This is expected because it did not have any -learned features to build on.

EfficientNetB0 did the worst. Since, its confusion matrix shows that it mostly predicted only one class for all the inputs indicating it did not learn patterns.

C. Grad-CAM and Clinical Interface

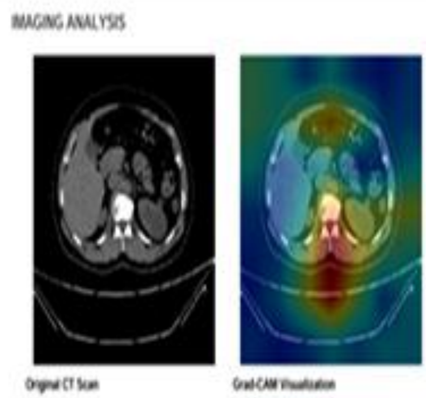
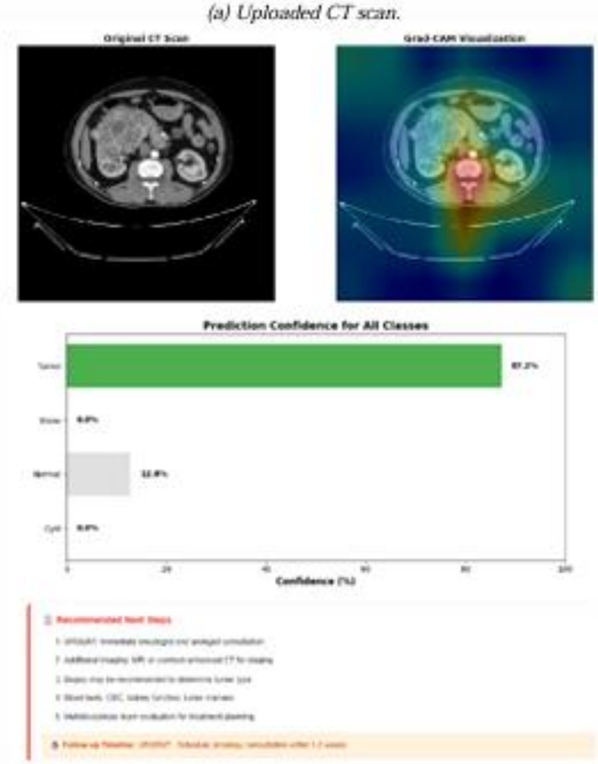


Fig. 7. *Showing what Grad-CAM visualization looks like for a kidney stone prediction*
The highlighted regions, which appear warm match the kidney area where stones are usually found indicating that the model is looking at the right spot for kidney stones.

Inspecting the Grad-CAM results I see that ResNet50 always pays attention to the parts of the body. For instance ResNet50 highlights the collecting system when it looks at stones and cysts and it looks at the outer kidney area, which is called the cortex, when it looks at tumors. The Grad-CAM results for ResNet50 show the things that doctors who look at pictures of the body called radiologists usually look for. This is really important. It means that ResNet50 is actually learning things that're useful for doctors like what ResNet50 looks for in pictures of the body rather than just looking at things that do not mean anything or at the background of the pictures.

(a) Uploaded CT scan.



(b) Predicted class and confidence.

(c) Report

(d) AI clinical assistant output.

Fig. 8. Clinical prediction interface for a Tumor case (87.2% confidence).

The Stone case in Fig. 8 shows how the system actually works. The model predicts Tumor with a high confidence of 87.2%. However, it is still labelled as Moderate Risk, because even when the model is very sure a doctor should always look at it before any treatment decisions are made.

The AI assistant then provides suggestions based on treatment for kidney stones. It seems sure, about its prediction indicating that the chances are:

Normal: 12.8%, Cyst: 0.0%, Stone: 0.0%, Tumor: 87.2%. The model is pretty confident. Still the system suggests a proper medical check. The kidney stone diagnosis is not something to take lightly.

D. Comparison of models

Reference	Model	Accuracy	Dataset	Explainability
Subedi et al. [5]	Vision Transformer	99.64%	CT Kidney (Kaggle)	No
Pimpalkar et al. [10]	InceptionV3	99.96%	CT Kidney (Kaggle)	No
Alzu'bi et al. [9]	2D-CNN	97.70%	Custom (KAUH)	No
Mehedi et al. [7]	VGG16 + Seg.	99.48%	CT Kidney (Kaggle)	No
Ahmed et al. [12]	VGG16 + Grad-CAM	97.41%	KUB X-ray	Yes
Proposed System	**ResNet50 + Grad-CAM + AI Assistant**	**81.04%**	**CT Kidney (Kaggle)**	**Yes (Full Suite)**

Table V. Comparison of this method with other existing models.

Some previous studies say that they got high accuracy above 99% but we need to be careful when we look at these results. Most of these studies used the Kaggle dataset and did not use any other external validation to check their results. Sometimes the way they split the data might have allowed the training data and the testing data to overlap. So just because a model gets accuracy it does not mean it will work well with new data from different hospitals or different scanners.

In our study we got an accuracy of 81.04%, which's a more realistic and reliable result under the given conditions. What is more important is that our study does more than just classification. Our study combines prediction with explainability, scores how confident we are, provides a clinical assistant, and automates reporting.

The study that is, closest to ours is the one done by Ahmed et al. [12], who also used Grad-CAM with VGG16. However, they used X-ray images and not CT scans, they also did not include a way to help doctors to make decisions based on how confident they are or give them any clinical recommendations.

VIII. DISCUSSION

A. Why ResNet50 Won

The ResNet50 model works better than the rest of the models because of its architecture as compared to the remaining ones. The special connections in ResNet50 help fix a problem that happens when a network gets too complicated. This problem is called the vanishing gradient problem which allows the network to go deeper (up to 50 layers) without losing important learning signals.

Because of this depth, the model can learn features at different levels. The model first starts to learn basic things like edges then it learns to see complicated things like shapes and structures.

This is really important when we are trying to tell the difference between kidney cysts, tumors and stones, where small differences in texture, density, and shape help make out the difference between them clearly.

ResNet50 also performed well in terms of precision by scoring about 0.865. This matters a lot in hospitals and medical settings since we cannot afford to tell someone they are sick when they are actually fine.

B. Why EfficientNetB0 Did Not Work

The performance of EfficientNetB0 was very low at first, but on looking at it more closely, it makes sense. EfficientNetB0 is trained for RGB (colored images) natural images. Hence, when you use it for grayscale (black and white) CT scans images which are quite different in terms of structure and information, the pre-trained features do not transfer well.

As a result, the model does not just perform slightly worse, it failed to learn useful patterns altogether. This shows that just because EfficientNetB0 does well on ImageNet it does not mean it will do well on medical images. For jobs, like this EfficientNetB0 needs to be tested and checked with the kind of data it will be used for, not just with natural image data.

C. Clinical Positioning

This system is made to help doctors to make decisions and for patients to get a faster review of their condition. However, it does not aim to replace the judgement of doctors rather act as an help. The Grad-CAM pictures help doctors understand why the system made a prediction. This way doctors can check the prediction before they do anything and also flags wrong predictions.

The clinical assistant part of the system helps doctors who are not specialists in a particular field. These doctors often need to check for kidney problems before a specialist doctor is available to see the patient. The system gives these doctors the information they need. The system tells doctors what a certain condition usually involves what symptoms they should look out for and what they should do next.

This is especially helpful in areas where it's hard to find specialist doctors and have limited resources but high demand of patients.

D. Limitations

There are some limitations of the work that we need to think about.

First, the testing for this work was done using a dataset from one place so we do not know how well the model will work with data from other hospitals or scanners or in different situations.

Second, the system only looks at one image at a time and does not consider the other factors of patients such as medical history or lab results or other information that doctors normally use to make a diagnosis.

The model also does not give any information about shape of the kidney lesion or size regarding how big it is, which are important things to know when we are planning a treatment.

Lastly, the AI assistant gives people more knowledge about their condition. This advice is the same for everyone with that condition. It does not give advice that's just, for that one person.

E. Future Work

There are a few things which we can do to make this system better:

We need to test the system in hospitals and with scanners to see how it really works.

We should add more details in the system to help it know more about the patient, like their age, their weight and lab results so we can have a deeper knowledge about their condition.

Which can help the system make more accurate predictions.The system has to be trained on data, from hospitals, while keeping the patient's information private.

IX.CONCLUSION

This paper presents the design, implementation, and evaluation of a complete Kidney Lesion Classification System using CNNs with transfer learning. The system focuses on four classes in CT images: Normal, Cyst, Stone, and Tumor.

A comparison of four models on the dataset showed that ResNet50 did the best with an accuracy of 81.04%. The final decision factors showed that it had a precision of 0.865, a recall of 0.810 and an F1-score of 0.819.

On the other hand, EfficientNetB0 performed very poorly, it had an accuracy of 20.23%, which is close to random guessing.This indicates that models that work well on pictures do not always do well with medical images. You need to test models, on the medical data to be sure.

What makes this system different from previous systems is not just its accuracy.It also does a lot of important things combining it all in one pipeline. Grad-CAM helps explain how it makes decisions.There is an assistant that can help not trained doctors and pateints to figure out what to do next. It can also make reports automatically making it easier to understand.

Each part of this system helps solve a problem that doctors face. The model has explainability segments which helps doctors understand what the system is thinking. The risk level's part points out when and how much the system is not sure about something.

This system needs better improvement before it can be used in medical field with full confidence. Most importantly, it needs to know more about the patients it is looking at. In addition, it needs to be tested on data from different hospitals and scanners, focusing more on patient-related information, adding features like lesion segmentation. However, as a proof, the system shows a clear and approach to building an AI tool which is not only accurate but also useful and understandable in medical field.

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